

An Lam

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EDUCATION

Master of Science in Computer Science, <i>University of Texas at Arlington</i>	GPA: 4.00 - Dec 2026
Bachelor of Science in Computer Science, <i>Texas Tech University</i>	GPA: 3.96 - Dec 2022

SKILLS

Machine Learning & AI	Classification & Regression, Linear Models (Regularization), Support Vector Machines (SVM), k-NN, Clustering, Neural Networks, Probabilistic Modeling (MLE, MAP), Model Evaluation, Search Algorithms (BFS, DFS, A*, Greedy), Game Playing
Mathematical Foundations	Linear Algebra, Probability & Statistics, Optimization (Gradient Descent)
Data & Systems	SQL, PostgreSQL, ETL Pipelines, Exploratory Data Analysis (EDA), Feature Engineering, Dimensionality Reduction, Text Data Modeling
Programming	Python (Pandas, NumPy, scikit-learn, TensorFlow), C/C++, JavaScript, SQL
Tools & Platforms	Docker, React, Node.js, Tableau

WORK EXPERIENCE

Database Administrator PostgreSQL, Python, React, Selenium	Jan 2023 — Mar 2025
<i>Phuong Tran Inc</i>	<i>Mentor, Ohio</i>
- Designed a PostgreSQL data warehouse with automated ETL pipelines to support BI reporting. - Built Python Selenium web scrapers to monitor competitor pricing and enable dynamic pricing strategies. - Developed a React + Node.js scheduling system, reducing idle time by 1 hour/day and increasing revenue by 15%. - Maintained POS systems and IT infrastructure, troubleshooting hardware, networking, and operational software issues.	
Student Researcher C/C++, Bash, RISC-V	Jan 2022 — Dec 2022
<i>Texas Tech University</i>	<i>Lubbock, Texas</i>
- Built automated benchmarking pipelines in Bash for large-scale RISC-V architecture experiments. - Developed visualization dashboards to identify compiler performance bottlenecks.	

PROJECTS

AI - 8 Puzzle Solver with Search Algorithms	Feb 2026
<i>Python, State Space Search, BFS, DFS, IDS, DLS, UCS, Greedy, A*</i>	
- Developed a Python-based state-space search solver for the classic 8-Puzzle problem, implementing BFS, DFS, DLS, IDS, UCS, Greedy Best-First, and A* search. - Designed and evaluated heuristic functions including Manhattan (L1) distance to guide A* search, significantly reducing node expansion and improving solution efficiency. - Performed algorithmic benchmarking comparing time complexity, memory consumption, and solution optimality across seven search strategies. - Implemented inversion-count based solvability detection and optimized state representation using hash-based visited-state tracking.	
Vector Space Search Engine	Feb 2026
<i>Python, Information Retrieval, TF-IDF, Cosine Similarity, NLP</i>	
- Built a vector-space information retrieval system using TF-IDF vectorization and cosine similarity for ranking document relevance to search queries. - Implemented scalable text preprocessing pipelines including tokenization, stemming, stop-word filtering, and normalization. - Evaluated retrieval effectiveness through ranking experiments and similarity-based document scoring.	
Portfolio Website with AI Assistant	Jan 2025
<i>React, Tailwind CSS, OpenAI API, Semantic Search, AWS, Docker</i>	
- Developed a full-stack portfolio platform integrating a semantic search engine using cosine similarity for contextual project discovery. - Built an AI-powered recruiter assistant using the OpenAI API to retrieve and summarize structured resume and project information. - Applied prompt engineering techniques to optimize LLM responses while minimizing token consumption and API cost. - Containerized and deployed the application using Docker on AWS with automated CI/CD pipelines.	
Biomedical Signal Classification	Mar 2024
<i>Python, Time-Series Analysis, HMM, Random Forest, Signal Processing</i>	
- Conducted exploratory analysis on MIT-BIH ECG time-series signals to identify discriminative patterns for cardiac rhythm classification. - Designed feature engineering pipelines using sliding-window segmentation and FFT-based frequency features. - Trained and evaluated Hidden Markov Models (HMM) and Random Forest classifiers, achieving 96% classification accuracy. - Compared model performance using cross-validation and statistical evaluation metrics.	